

TRADEPULSE

Report A — Rubric-Aligned Application Pack

Documentary trade-compliance decision support for bank and GIFT IFSC trade-house officers
Track 1 — Agentic AI | Cross-border trade finance

Technical execution (mandatory requirements, working live)	25%	\$1 Application / POV / Infra / Architecture
Problem clarity and depth	15%	\$2
Solution Innovation & Differentiation	20%	\$3
Market Viability	15%	\$4
Presentation / demo clarity	15%	\$5
Team Assessment (Psychometric) — Round 1 only	10%	\$6

*Note: Official guidebook lists Technical execution at **25%** (not 30%). This pack still treats technical depth as the primary deliverable for judges.*

Live demo (hackathon): Web + API on AWS ECS Fargate + ALB, region ap-south-1

Product stance: decision-support only — never autonomous approval, sanctions confirmation, or Customs clearance.

1. Technical execution — Application report, POV, infra, architecture

Point of view. Trade banks and IFSC trade houses still examine documentary packs under UCP 600–style discipline: extract fields, reconcile invoice vs transport, check identity and sanctions candidates, and leave a maker–checker trail. Generic “AI document checkers” collapse fuzzy name matches into false certainty and hide provenance. TradePulse is built as **examiner decision support**: multi-agent extraction with bounded debate, deterministic policy, explicit failure states (DATA_UNAVAILABLE → never PASS), and an identity confidence ladder that separates LEI candidates from vLEI verification.

1.1 What we shipped (mandatory / live)

- End-to-end case flow: create case → upload invoice (+ BoL for post-shipment) → process → workbench review.
- Agentic document intelligence: Extractor → Validator → Challenger → Arbiter → Cross-document reconciler (max 3 debate rounds).
- Identity ladder API + UI: LEI / name-match / vLEI fixture states with REVIEW_REQUIRED when evidence is weak.
- Examiner case pack download for human review handoff.
- Maker–checker style workbench language; no “AI approved / cleared / sanctioned” claims.
- Working live on AWS: public ALB URLs for web and API (ap-south-1).
- Synthetic demo credentials labeled; mock/fixture adapters where live regulated APIs are not configured.

1.2 System architecture (high level)

Monorepo: **apps/web** (Next.js workbench) → **apps/api** (FastAPI) → adapters (Bedrock, Textract, S3, GLEIF, sanctions) → typed contracts in **packages/contracts**. Policy and rule packs are versioned; audit is append-only; replay must not overwrite history.

Presentation	Marketing + examiner workbench, case queue, identity ladder, Next.js typescript	
API / orchestration	Cases, uploads, process pipeline, identity ladder, examiner pack	FastAPI, Pydantic
Agent swarm	Bounded roles; evidence-cited debate; arbiter cannot invent facts	Bedrock LLM adapters
Document intake	OCR / text extraction from invoice & transport docs	Amazon Textract / parsers
Identity & lists	GLEIF LEI candidates; vLEI fixture/live adapter; sanctions candidates	GLEIF adapters + policy
Storage & secrets	Object store for docs; env/secrets for keys; optional Dynamo/locks	S3, IAM, ECS task env
Hosting	Containerized API + web behind ALBs	ECR, ECS Fargate, ALB

1.3 AWS services used (hackathon deployment)

Amazon ECS (Fargate)	Runs API and Next.js standalone containers without managing EC2
Elastic Load Balancing (ALB)	Public HTTP entry for web and API; health checks
Amazon ECR	Stores Docker images for API and web
Amazon VPC / subnets / SG	Network isolation; ALB → tasks; egress for AWS APIs
Amazon S3	Document object storage for uploaded trade packs
Amazon Textract	OCR / structured extraction assist on PDFs/images
Amazon Bedrock	LLM backbone for agent roles (extraction, challenge, arbiter summaries)
IAM roles / task roles	Least-privilege access to S3, Textract, Bedrock
CloudFormation (deploy scripts)	Repeatable stacks: api-ecs / web-ecs

CloudWatch (logs)	Container stdout/stderr for demo debugging
(Deferred) Amplify Hosting	Optional web host; deferred — needs GitHub OAuth; ECS used instead

Region: **ap-south-1** (Mumbai). Account used for demo deploy: team AWS profile *tradepulse*. Live pattern: Web ALB → Next.js; browser calls API ALB with CORS origin allowlist.

1.4 Hosting topology & URLs

- Web: ECS service behind ALB (example pattern: *tradepulse-web-*.ap-south-1.elb.amazonaws.com*).
- API: ECS service behind ALB (example pattern: *tradepulse-api-*.ap-south-1.elb.amazonaws.com*).
- CORS: API allows the web ALB origin; empty `AWS_PROFILE` stripped in ECS so Bedrock/Textract use task IAM role.
- Local alternative: API :8000 + web :3000 for development; same contracts.

1.5 Infra cost envelope (hackathon / always-on estimate)

Rough always-on cost in ap-south-1 for the current demo shape (2 Fargate services + 2 ALBs + light S3/Textract/Bedrock usage) is on the order of **USD ~40–60 / month** if left running 24/7. Variable spend is dominated by Bedrock tokens and Textract pages during demos. Tear-down: delete CloudFormation stacks when idle to stop ALB/Fargate burn.

ECS Fargate (API + Web)	vCPU · memory · hours	Small tasks; tear down after demos
Application Load Balancers (×2)	Hourly ALB + LCU	Largest fixed cost; share ALB later
ECR storage	GB-month of images	Negligible at demo scale
S3	GB + PUT/GET	Low for fixture packs
Textract	Pages analyzed	Spike during live demo uploads
Bedrock	Input/output tokens	Spike during process pipeline
Data transfer	ALB egress	Low for jury traffic

1.6 Scale-up path (post-hackathon)

- Single shared ALB + path-based routing; or CloudFront + ACM HTTPS custom domain.
- Async job queue (SQS + worker tasks) for long Textract/Bedrock pipelines; API returns case status.
- Aurora/Postgres or DynamoDB for multi-tenant case store; today demo may use local/in-memory patterns.
- WAF on ALB; Secrets Manager for non-IAM credentials; private subnets for tasks.
- Separate staging/prod accounts; rule-pack promotion with human approval only.
- Horizontal scale: more Fargate tasks on CPU/memory alarms; Bedrock provisioned throughput if needed.
- Observability: structured audit events + CloudWatch metrics/alarms on 5xx and pipeline latency.

1.7 Safety & non-goals (technical honesty)

- Not a Customs portal, ICEGATE filer, payment engine, or container inspection system.
- Fuzzy LEI/name match ≠ identity proof; potential sanctions hit ≠ confirmed match.
- `DATA_UNAVAILABLE` / `NOT_AVAILABLE` must never become `PASS`.
- Agent consensus is extraction confidence only — never legal/compliance finality.
- vLEI fixture = `SYNTHETIC_DEMO_CREDENTIAL` unless a trusted live verifier is configured.

2. Problem clarity and depth (15%)

Problem. Cross-border documentary trade finance still depends on humans reconciling noisy PDFs under time pressure. False positives on sanctions/name screening and silent data gaps create either (a) rubber-stamp risk or (b) endless exception queues. GIFT City IBUs and trade houses need examiners who can move faster *without* pretending the model is the compliance officer.

Depth we claim (and evidence)

- Document policy awareness: Commercial Invoice required; BoL/AWB conditionally required by profile; distinguish REQUIRED vs NOT_AVAILABLE.
- Cross-document reconciliation: invoice vs transport fields with explicit unresolved states.
- Identity ladder: GLEIF candidates, LEI status warnings (e.g. lapsed ≠ fraud), separate vLEI evidence.
- TBML-adjacent substance signals (price/qty anomalies) as review cues — not fraud verdicts.
- Audit: versions, provenance (source/snapshot where applicable), agent disagreement visible.
- Buyer language: Head of Trade Finance Ops / Transaction Banking — not “all banks and fintechs.”

Why this team / why this problem. We scoped ownership (backend platform, workbench product, UI quality, QA) and locked contracts so the demo cannot invent enums mid-pitch. The problem is owned as examiner workflow + evidence discipline, not as “another chat PDF summarizer.”

3. Solution Innovation & Differentiation (20%)

Commodity LC/document AI tools sell speed. TradePulse differentiates on **epistemic honesty** and examiner UX:

Identity	Fuzzy name → “matched”	Ladder: candidate → LEI-supported → vLEI-supported; else REVIEW
Sanctions	Hit = blocked story	Candidate + policy; no auto-confirm without authority
Agents	One LLM prompt	Bounded swarm + arbiter; max 3 rounds; disagreements shown
Missing data	Silent skip / soft pass	Typed unavailable states; cannot PASS
Output	Approve/reject badge	Examiner pack + maker–checker language
Infra story	Laptop demo	Live AWS ECS + real Bedrock/Textextract path

Innovation is not “more agents.” It is encoding bank-grade failure modes into the product so the jury can trust the demo’s claims.

4. Market Viability (15%)

4.1 Customer specificity

First paying persona (example judges expect): Head of Trade Finance Operations at a GIFT City IBU (or mid-size private bank trade desk) who owns exception queues, examiner productivity, and audit findings — not a generic “fintech.” Economic buyers also include Transaction Banking heads and compliance leaders who care about maker–checker evidence packs.

4.2 Why now

- Agentic AI cost curves make multi-step document debate affordable vs 2022-era single-shot OCR.
- LEI/vLEI and organisational identity rails are maturing for cross-border counterparties.
- IFSC / GIFT City trade finance growth increases documentary volume under regulated scrutiny.
- Banks face pressure to cut TAT without increasing model-risk findings — decision support fits better than autopilot.

4.3 GTM sketch

- Land: pilot with 1 corridor + invoice+BoL profile; measure exception reduction and audit completeness.
- Expand: more document types per policy; SSO; on-prem/VPC deploy option.
- Monetise: per-case + seat SaaS or bank VPC license; professional services for rule-pack onboarding.
- What we would NOT sell week one: autonomous approval, live Customs filing, unlabeled synthetic credentials as real.

5. Presentation / demo clarity (15%) — with example script

Judges score clarity of demo, not slide density. Use one narrative: **upload** → **agents disagree** → **policy holds** → **human decides**.

5.1 Recommended 4-minute demo arc

- 0:00–0:30 — Problem: examiner drowning in PDFs; AI tools overclaim. TradePulse = decision support.
- 0:30–1:00 — Open live URL. Show case queue. Create/open a fixture case with invoice + BoL.
- 1:00–2:00 — Processing rail: agents running. Call out max 3 rounds and evidence requirement.
- 2:00–2:45 — Workbench: field with REVIEW_REQUIRED; identity ladder (candidate ≠ verified).
- 2:45–3:30 — Download examiner pack; show maker cannot be skipped by “AI consensus.”
- 3:30–4:00 — Safety line: DATA_UNAVAILABLE never PASS; not Customs; not autonomous sanctioning.

5.2 Example spoken lines (copy-ready)

“We’re not asking the model to approve the trade. We’re asking it to propose fields, challenge itself, and leave an evidence pack a Head of Trade Ops can defend in audit.”

“This GLEIF hit is a candidate. Without a matching LEI on the document, the ladder stops at potential match — REVIEW_REQUIRED. That’s intentional.”

“If Textract or Bedrock is down, you see SOURCE_UNAVAILABLE — not a green pass. Missing data is the failure mode banks actually fear.”

5.3 Visual hygiene

- One browser tab with live ALB URL; backup: localhost recording if network fails.
- Pre-warm one case so demo isn’t waiting on cold Bedrock.
- Avoid claiming Hybrid/Quantum tracks; stay Track 1 Agentic AI.
- Label synthetic vLEI fixture on screen if shown.

6. Team Assessment — Psychometric (Round 1 only, 10%)

Round 1 includes a team psychometric assessment. This section is a preparation brief — not a fake “test answer key.” Panels look for coachability, role clarity, conflict handling, and whether ownership matches the pitch.

Role clarity	Backend/platform & contracts; web product/workbench; UI/UX polish; QA/release — named owners, not “everyone”
Decision style	Contracts + PRD + system design as authority; stop on conflict rather than invent enums mid-demo.
Conflict	Disagreement on shared contracts → no silent merge; short review then ADR.
Coachability	If R1 feedback says “narrow the buyer,” R2 deck names Head of Trade Ops at GIFT IBU — not “banks.”
Stress / load	Demo failover plan (prebuilt case); infra tear-down discipline; honesty about what breaks next week.
Ethics	Refuse to demo “AI cleared the shipment”; refuse unlabeled synthetic as live regulatory truth.

Practical tips for the psychometric / team interview

- Answer as “we” with specific “I own X” examples — avoid both hero-solo and vague collective.
- Admit one real trade-off (e.g. Amplify deferred; ECS chosen for speed) and what you’d change Monday.
- Show you can take feedback: cite one change made after internal challenge (identity ladder, CORS/IAM fix, etc.).
- Don’t oversell completeness; judges reward teams who know prototype vs product.

End of Report A. Pair with Report B for judge Q&A; fire drills.